POSTS AND TELECOMMUNICATIONS
INSTITUE OF TECHNOLOGY
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INTERNSHIP

REPORT

WoxionChat

Name: NGUYEN PHAM TRUNG HIEU
Student ID: B23DCCE031
Class ID: B23CQCE04-B
Instructor: KIM NGOC BACH

Hanoi

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1 Overview

This week marked the successful completion of the core development phase for both the **Backend** and **Frontend** of the WoxionChat project. Our primary focus was on bridging any remaining functional gaps, optimizing system communication, and performing comprehensive end-to-end (E2E) testing to ensure production readiness.

2 Technical Accomplishments

2.1 Backend Finalization

The Django-based infrastructure reached its final stable version for the initial release:

- **API Polish:** Refined all REST endpoints to handle edge cases in user input and AI agentic reasoning loops.

2.2 Frontend Completion

The React-based user interface was fully integrated with the backend services:

- **Full Feature Integration:** Completed the synchronization between the chat interface and the AI API, including real-time streaming and interactive tool-calling visualizations.
- **State Management Refinement:** Optimized Redux store performance to handle large datasets and complex session states without UI stuttering.
- **Final UI/UX Polish:** Conducted a comprehensive design review to ensure visual consistency, accessibility (a11y), and responsiveness across all breakpoints.
- **Error Handling UI:** Implemented user-friendly error boundaries and feedback notifications for network interruptions or AI processing timeouts.

2.3 System-Wide Integration & Testing

- **End-to-End (E2E) Testing:** Successfully executed a full suite of automated tests covering the entire user journey, from registration and login to complex agentic AI interactions.
- **Performance Benchmarking:** Verified that the integrated system meets the target latency requirements

3 Technology Stack Finalization

- **Backend:** Django, Django Channels, Celery, Redis.
- **Frontend:** React, Tailwind CSS, Lucide-React.
- **Database:** MongoDB (NoSQL) with optimized indexing.
- **Testing:** pytest (Backend).

4 Progress Evaluation

- **Status: Feature Complete**.
- **Observations:** The seamless integration of the "The Boss" dashboard with the AI-driven employee monitoring tools has received positive internal feedback. The system is stable and ready for staging deployment.

5 Upcoming Milestone (Final Phase)

1. **Production Staging:** Deploying the full stack to the production environment for final smoke testing.
2. **Project Documentation:** Finalizing technical manuals, API documentation, and user guides for the handover.
3. **Final Stakeholder Demo:** Presenting the completed WoxionChat platform to project sponsors.